

~\OneDrive - Florida State University\FA25\Mech2\Project\Arduino\main\main.ino

```
1 #include "src/myStepper/myStepper.h"      // doing it like this so git will keep up
2 #include "src/myDCEncoder/myDCEncoder.h"  // doing it like this so git will keep up
3
4 // Change this so that all these pins are definitions up top
5 // Two encoders
6 #define motorA_IN1 5
7 #define motorA_IN2 6
8 #define motorA_EN 9    // pwm pin
9 #define motorA_chA 2    // encoder
10 #define motorA_chB 3    // encoder
11
12 #define stepperIN1 40
13 #define stepperIN2 41
14 #define stepperIN3 42
15 #define stepperIN4 43
16
17 myDCEncoder motorA(motorA_IN1, motorA_IN2, motorA_EN, motorA_chA, motorA_chB);
18
19 myStepper mS(stepperIN1, stepperIN3, stepperIN2, stepperIN4);
20
21 int desPos1 = 400;
22 int desPos2 = -400; // roughly 180 degrees
23 int desPos = desPos1;
24
25 int direction = 1;
26 bool state = 0;
27
28 void setup() {
29
30     mS.init();
31     motorA.init();
32
33     attachInterrupt(digitalPinToInterrupt(motorA_chA), ISR_motorA_chA, CHANGE);
34     attachInterrupt(digitalPinToInterrupt(motorA_chB), ISR_motorA_chB, CHANGE);
35
36     Serial.begin(9600);
37     delay(2000);
38 }
39
40 void loop() {
41
42     switch (state){
43         case 0:
44             if (abs(motorA._curPos - desPos1) < 8){
45                 state = !state;
46                 desPos = desPos2;
47             }
48             break;
```

```
49
50     case 1:
51     if (abs(motorA._curPos - desPos2) < 8){
52         state = !state;
53         desPos = desPos1;
54     }
55     break;
56
57
58 }
59
60 motorA.PID(desPos);
61
62 Serial.println(motorA._curPos);
63 }
64
65 void ISR_motorA_chA() {
66     motorA.updateCurState();
67 }
68 void ISR_motorA_chB() {
69     motorA.updateCurState();
70 }
```